CS 391
Web Application Development

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Week - 2

Introduction to CSS

CSS Basics

1. Introduction to CSS
2. CSS Basic Selectors
3. Cascading Order
4. CSS Formatting

What is CSS?

- CSS : Cascading Style Sheets
  - Describes how HTML elements will be displayed on the screen, paper, or in other media.
  - Saves a lot of work. It can control the layout of multiple web pages all at once.
- External stylesheets are stored in CSS files.

CSS Solved a Big Problem

- Development of large websites become a nightmare for web developers because of using tags for styling such as <font>
- To solve this problem, the World Wide Web Consortium (W3C) created CSS.
- CSS removed the style formatting from the HTML page!

CSS Benefits …

- Site wide changes are easily implemented
  - HTML is easier to develop and maintain
  - HTML will work in future browsers.
- Better user experience
- Shorter Development Time = Faster time to market
- Smaller Pages = Less Bandwidth
  - Page sizes can be reduced by 25-50%

...CSS Benefits

- Without nested tables, pages are rendered faster by the browser.
- With an external stylesheet file, you can change the look of an entire website by changing just one file!

CSS Syntax

- The CSS syntax is made up of three parts:

SELECTOR PROPERTY VALUE

```
p { color: blue; }
```

```
p {
      text-align: center;
      color: black;
      font-family: arial
}
```

```
h1,h2,h3
{
    color: red
}
```

Inserting a style sheet

- Three ways of inserting a style sheet
  - External Style Sheet
  - Internal Style Sheet
  - Inline Styles
- External style sheets should be used

External Style Sheet

- An external style sheet is ideal when the style is applied to many pages
- Each page must link to the style sheet using the <link> tag.
- Style sheet file:
  - should be saved with a .css extension
  - should not contain any html tags

This is first paragraph

This is second paragraph

HTML file

```
<head>
   <link rel="stylesheet" href="styles.css" />
</head>

<body>
   <p class="first">This is first paragraph</p>
   <p class="second">This is second paragraph</p>
</body>
```

CSS file

```
p.first{
   text-align: left;
   color:red;
   font-size: 20px; }
p.second{
   text-align:left;
   color:blue;
   font-size: 16px; }
```

Internal Style Sheet

- Internal Styles
  - should be used when a single document has a unique style
  - define by using <style> tag in the head section

```
<head>
   <style>
      p {
         color: white;
      }
      body {
         background-color: black;
      }
    </style>
</head>
```

Inline Styles

- Placing CSS right in the HTML code
- This method should be used carefully
  - For example, when a style is applied to a single occurrence of an element.

```
<p style="background: black; color: white;">
  This is new background and font color with inline CSS
</p>
```

CSS Basic Selectors

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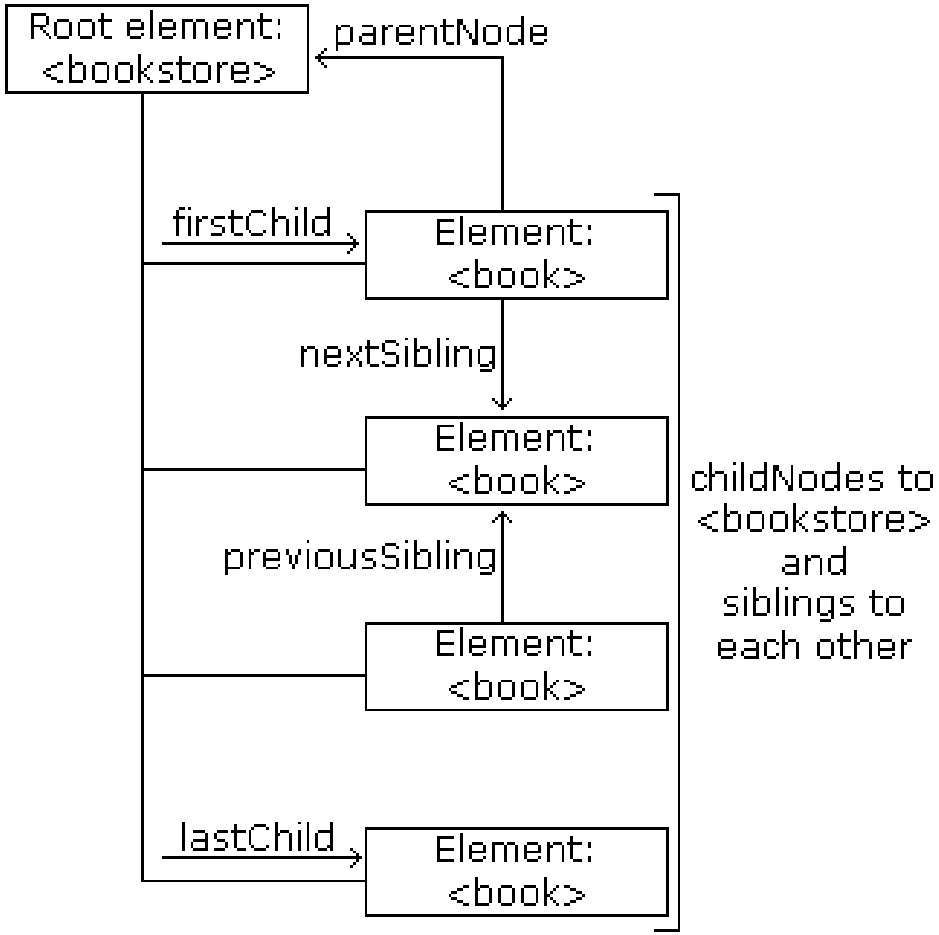
DOM Nodes


Root element:
<bookstore>
parentNode
firstChild
Element:
<book>
nextSibling
Element:
<book>
previousSibling
Element:
<book>
lastChild
Element:
<book>
childNodes to
<bookstore>
and
siblings to
each other

Selectors

- There are many different types of selectors
  - Type Selector
  - Class Selector
  - ID Selector
  - Descendant Selector
  - Universal Selector
  - Child Selector
  - Adjacent Sibling Selector
  - Attribute Selector
- Multiple selectors may be associated with same style rules
  - separate them by comma

CSS

```
h1,h2,h3
{
    color: red;
    font-family: arial
}
```

Type Selector

- A selector which selects elements in the (DOM)
- Ex:
  - h1
  - body
  - td
  - br
  - table
  - tr
  - …

CSS

```
p {
    text-align: center;
    color: red;
    font-size: 20px;
}
```

HTML

```
<p>Faculties</p>
<p>Faculty of Business</p>
<p>Faculty of Engineering</p>
```

RESULT

Faculties

Faculty of Business

Faculty of Engineering

Class Selector

- Applies a style to an element having the specified class attribute.
  - Specified with '.' before the class name
  - Examples
    - p.quote - Applies to all ***paragraphs*** with a class of "quote"
    - .error - Applies to any element with a class of "error".

HTML

```
<p class="first">Faculty of Business</p>
<p class="second">Faculty of Engineering</p>
```

CSS

```
p.first {
    text-align: left;
    color: red;
}
p.second {
    text-align: right;
    color: blue;
}
```

RESULT

Faculty of Business

Faculty of Engineering

...Class Selector

- HTML elements can also refer to more than one CSS classes
  - But only one class attribute can be specified per HTML element
- To define a style that will be used by all HTML elements
  - tag name can be omitted

CSS

```
.myclass1 {
     text-align: center;
     color:purple;
}

.myclass2 {
     font-weight: bold;
     font-size: 32;
     font-family:"Lucida Sans";
}
```

HTML

```
<p class="myclass1">Faculty of Business</p>
<p class="myclass1 myclass2">Faculty of Engineering</p>
```

RESULT

Faculty of Business

Faculty of Engineering

ID Selectors

- The id selector uses the id attribute of an HTML element to select a specific element.
  - Id must be ***unique*** in the page.
  - To select an element with a specific id, write a hash (#) character, followed by the id of the element

CSS

```
#business {
    text-align: center;
    color: blue;
}
```

HTML

```
<p id="business">Faculty of Business</p>
```

RESULT

Faculty of Business

The CSS Grouping Selector

- To group selectors, separate each selector with a comma.

```
h1 {
  text-align: center;
  color: red;
}

h2 {
  text-align: center;
  color: red;
}

p {
  text-align: center;
  color: red;
}
```

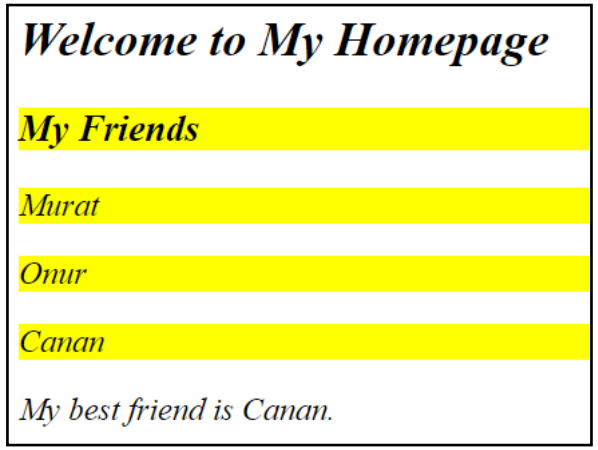
```
h1, h2, p {
  text-align: center;
  color: red;
}
```

It will be better to group the selectors, to minimize the code.

Universal Selector

- Used to select any element.
  - Acts as a wildcard symbol.

RESULT



CSS

```
* {
    font-style: italic;
}

div * {
     background-color: yellow;
}
```

HTML

```
<h2>Welcome to My Homepage</h2>

<div>
    <h3>My Friends</h3>
    <p>Murat</p>
    <p>Onur</p>
    <p>Canan</p>
</div>
<p>My best friend is Canan</p>
```

Some Rules

- HTML names should match the name of the selector exactly.
  - **Wrong**
    - <p class="red" /> does not match **p.Red{}**
  - **Correct**
    - <p class="red" /> matches **p.red{}**
- No spaces
  - **Wrong**
    - <input id="campus street" />
  - **Correct**
    - <input id="campus-street" />
    - <input id="campusStreet" />

Cascading Order

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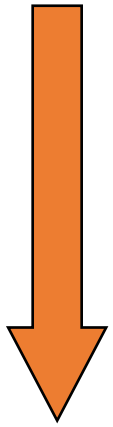
Cascading Order

- What style will be used when there is more than one style specified for an HTML element?
  - Inline Styles
  - Internal Style Sheets (Embedded)
  - External Style Sheets (Linked or Imported)
  - Browser Defaults



highest priority

lowest priority

Example-1…

HTML

```html
<html>
<head>
    <link rel="stylesheet" href="styles/mystyles.css">
    <style type="text/css">
        p {
            color: green;
        }
    </style>
</head>
<body>
    <p style="color: blue">Welcome</p>
</body>
</html>
```

mystyles.css

```css
p {
    color: red;
}
```

Which one wins ?

…Example-1

Inline styling wins → Welcome

Example-2…

HTML

```html
<html>
<head>
    <link rel="stylesheet" href="styles/mystyles.css">
    <style type="text/css">
        p {
            font-weight: bold;
            font-size: 18px;
        }
    </style>
</head>
<body>
    <p>Welcome</p>
</body>
</html>
```

mystyles.css

```css
p {
    color: red;
}
```

Which one wins ?

…Example-2

Both of them

Example-3…

HTML

```html
<html>
<head>
    <link rel="stylesheet" href="styles/mystyles.css">
    <style type="text/css">
        p {
            color: blue;
        }
        p {
            color: fuchsia;
        }
    </style>
</head>
<body>
    <p>Welcome</p>
</body>
</html>
```

mystyles.css

```css
p {
    color: red;
}
```

Which one wins ?

...Example-3

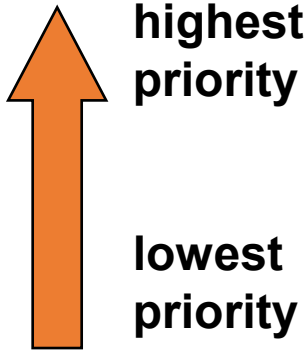
Internal styling wins → Welcome

<u>one basic rule:</u>

If two styles come into conflict, the last one used will take precedence

Selector Type Precedence

- Selector precedence occurs in this order
  - ID
  - Class
  - Element



Example…

HTML

```
<html>
<head>
    <link rel="stylesheet" href="styles/mystyles.css">
    <style type="text/css">
       #sample {
            color: blue;}
        p {
            color: green;}
        p {
            color: red;}
    </style>
</head>
<body>
    <p>Welcome</p>
    <p id="sample">My Sample Page</p>
    <p class="prg">Sample Paragraph</p>
</body>
</html>
```

mystyles.css

```
.prg{
     color:fuchsia;
}
```

Which one wins ?

...Example

Will use Internal style

Will use id selector

```
<p>Welcome</p>
<p id="sample">MySamplePage</p>
<p class="prg">Sample Paragraph</p>
```

Class selector will be used

Welcome

MySamplePage

Sample Paragraph

CSS Formatting

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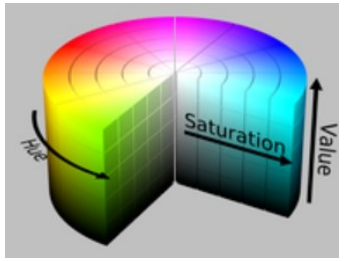
CSS Colors

- Colors are specified with predefined following ways:
  1. color names (HTML supports 140 standard color names)
  2. RGB - (red, green, blue)
     - Each parameter (red, green, and blue) defines the intensity of the color with a value between 0 and 255.
     - This means that there are 256 x 256 x 256 = 16.777.216 possible colors!
  3. RGBA - Red-green-blue-alpha
     - alpha channel - which specifies the opacity of the color
  4. HEX
     - #rrggbb -rr (red), gg (green) and bb (blue) are hexadecimal values between 00 and ff
  5. HSL-(hue, saturation, lightness)
     - Hue is a degree on the color wheel from 0 to 360. 0 is red, 120 is green, and 240 is blue.
     - Saturation is a percentage value, 0% means a shade of gray, and 100% is the full color.
     - Lightness is also a percentage value, 0% is black, and 100% is white.
  6. HSLA - (hue, saturation, lightness, alpha)



Color Examples…

```
<h1 style="background-color:blue;">Hello</h1>
<p style="background-color:red;">Web Programming</p>
<p style="color:blueviolet;">Formatting Samples</p>
```

	Black		Gray		Silver		White
	Yellow		Lime		Aqua		Fuchsia
	Red		Green		Blue		Purple
	Maroon		Olive		Navy		Teal

```
<h1 style="background-color:rgb(255, 80, 100);">Hello</h1>
<p style="background-color:rgb(0, 255, 98);">Web Programming</p>
<p style="color:rgb(0, 0, 255);">Formatting Samples</p>
```

Color	Color RGB
	rgb(0,0,0)
	rgb(255,0,0)
	rgb(0,255,0)
	rgb(0,0,255)
	rgb(255,255,0)
	rgb(0,255,255)
	rgb(255,0,255)
	rgb(192,192,192)
	rgb(255,255,255)

...Color Examples

```
<h1 style="background-color: #FFFF00">Hello</h1>
<p style="background-color: #FF00FF">Web Programming</p>
<p style="color: #FF0000">Formatting Samples</p>
```

Hello

Web Programming

Formatting Samples

Color	
	#000000
	#FF0000
	#00FF00
	#0000FF
	#FFFF00
	#00FFFF
	#FF00FF
	#C0C0C0
	#FFFFFF

CSS Background

- Defines the background effects on an element
  - **background**
    - all background properties in one declaration.
  - **background-attachment**
    - sets whether a background image is fixed or scrolls with the rest of the page.
  - **background-color**
    - background color of an element
  - **background-image**
    - Sets an image as the background
  - **background-position**
    - sets the starting position of a background image
  - **background-repeat**
    - sets if/how a background image will be repeated

CSS Background Example

```
body {
      background-image: url(background.jfif);
      background-repeat:repeat-x repeat-y;
}
```

Shorthand:

```
body {
    background: #ffffff url("background.png") no-repeat right top;
}
```

https://developer.mozilla.org/en-US/docs/Web/CSS/Shorthand_properties

order of the property values:

background-color
background-image
background-repeat
background-attachment
background-position

CSS Text

- Defines the spacing, decoration, and alignment of text
- Properties
  - color
  - direction
  - letter-spacing
  - text-align
  - text-indent
  - text-decoration
  - white-space
  - word-spacing

CSS Text Example

```
h1 {color:#00ff00;
    text-align:center;
    text-decoration:underline;}
h2 {text-decoration:line-through;}
h3 {text-decoration:overline;}

p.date {
   text-align:right;}
p.main {
   text-align:left;}

p.uppercase {
   text-transform:uppercase;}

p.lowercase {
   text-transform:lowercase;}

p {text-indent:50px;}
```

header 1

~~**header 2**~~

header 3

03/10/2003

Our customers have software related issues that are tied to complex businesses that are ever changing, requiring adaptability in order to stay competitive and maintain profitability. Organizations do not have the people who can see through the maze of information quickly, not only within their own organization but also externally with the many software systems they are integrated with.The complexity and cost of maintaining information is ever increasing.

SAMPLE PARAGRAPH

CSS Font

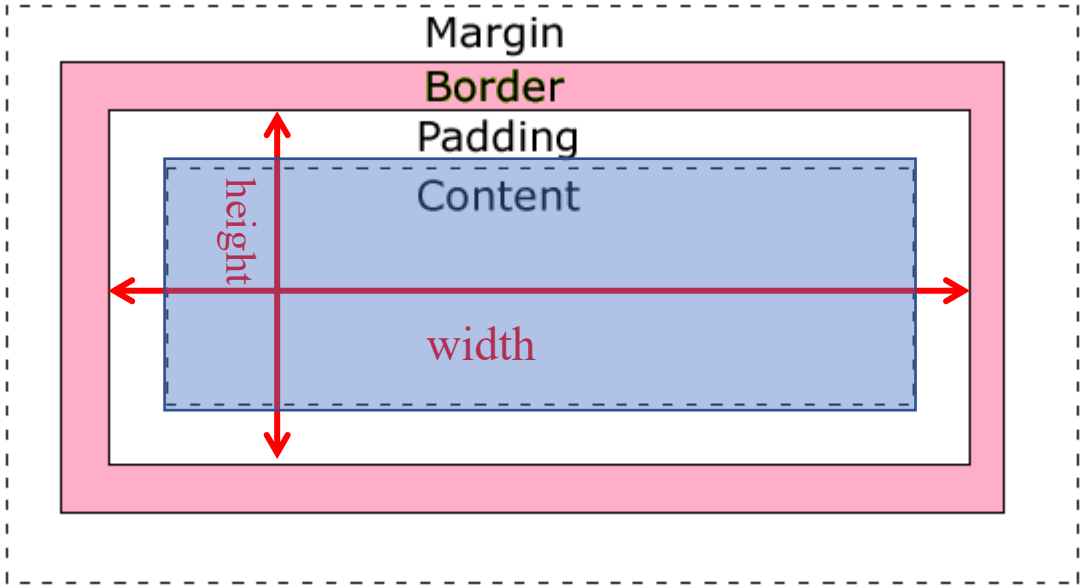
- Defines the font in text
- Properties
  - font
  - font-family
  - font-size
  - font-style
    - normal, italic, oblique
  - font-weight
    - normal, bold,...
  - ...

```
p {
  text-indent: 50px;
  font-family: 'Lucida Sans', 'Lucida Sans Regular’;
  font-weight: bold;
  font-size: 12;
  font-style: italic;
}
```

A dolphin that has lived in the sea near the town of Dingle, Ireland, for nearly 40 years has gone missing. After not being seen since mid-October, there are concerns that the dolphin, called Fungie, has died or left the area. When Fungie first came to Dingle in 1983, it was a very poor area and did not have a strong tourism industry. Caroline Boland from the Dingle Peninsula Tourism Alliance told The Journal that after Fungie arrived, people started traveling there to see him.

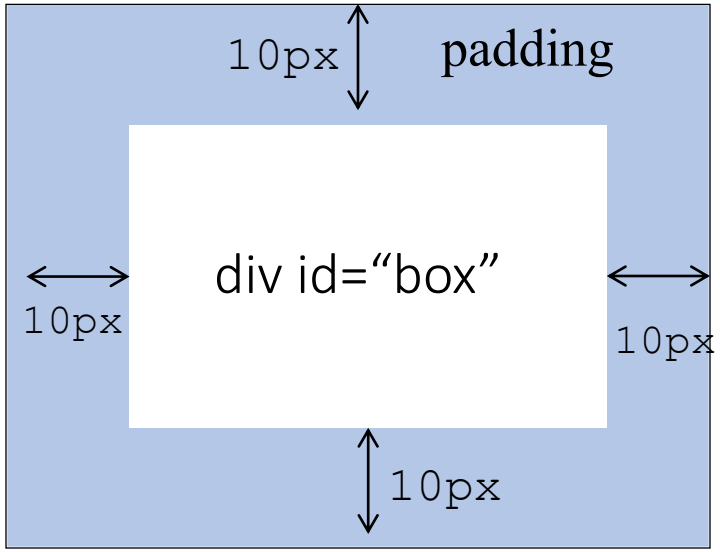
CSS Box Model

- All HTML elements can be considered as boxes.
  - box wraps around HTML elements, and it consists of:
    - margins, borders, padding, and the actual content.



Padding

- Padding is the space between the text/content and the border.

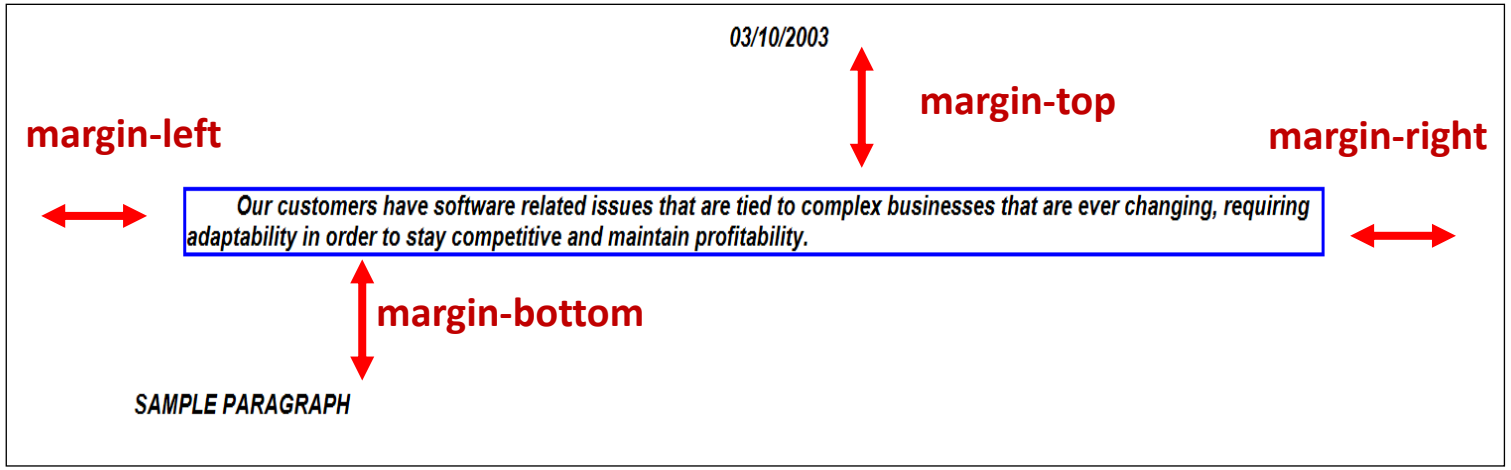



```
padding-left: 10px;
padding-right: 10px;
padding-bottom: 10px;
padding-top: 10px;
```

CSS Margin

- Defines the space around the elements
- Properties
  - margin
  - margin-bottom
  - margin-left
  - margin-right
  - margin-top

CSS Margin Example




```
h2{
    margin-top: 100px;
    margin-right: 100px;
    margin-bottom: 100px;
    margin-left: 100px;
    border: 3px solid blue;
}
```

CSS Border

- Allows for complete customization of the border that appear around HTML elements
- Properties
  - border
  - border-color
  - border-style
  - border-bottom
  - border-bottom-color
  - border-bottom-style
  - border-bottom-width
  - ....

CSS Border Example

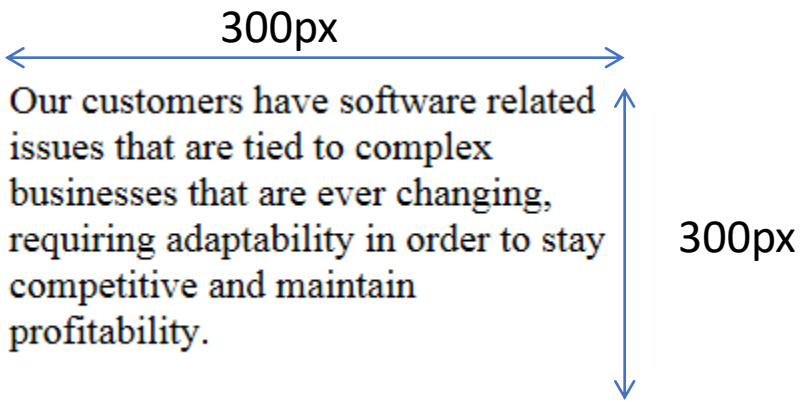
```
table {
    border-width: 3px;
    border-style: double;
    border-bottom-color: red;
    border-top-color: blue;
}
td {
    border-width: thin;
    border-style: dashed;
}
p {
    border-width: thick;
    border-style: solid;
}
```

Welcome to My Homepage

Esma	Meral
Özlem	Köz
Elvan	Çelik

Width & Height

- Width and height define the width and height of an element




```
p.box
{
    height:300px;
    width:300px;
}
```

Box Sizing

sets how the total width and height of an element is calculated.

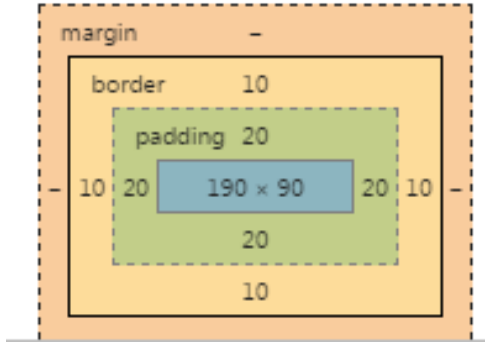
Default:
width + padding + border = actual width
height + padding + border = actual height

CSS

```
#box {
    width: 250px;
    height: 150px;
    border: 10px solid black;
    box-sizing: border-box;
    padding:20px;
}
```

Actual width=250

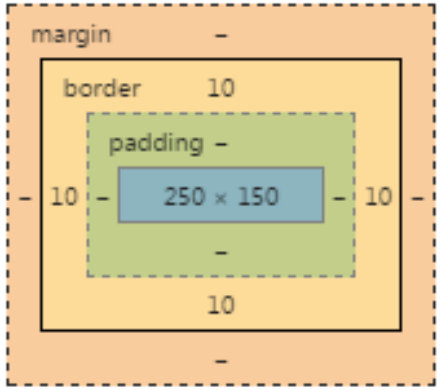
Content width=190
(250-30-30)



CSS

```
#box {
    width: 250px;
    height: 150px;
    border: 10px solid black;
    box-sizing: content-box;
}
```

Actual width=270
(250+10+10)



What You Have Learned?

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Any Question?

Thank you for your attention

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